Loan Approval Prediction

Predicting whether a loan application will be approved.

Problem

This project predicts whether a loan application will be approved, using details about the applicant such as income, loan amount, credit history and property area.

Data and tools

I used the Kaggle **Loan Prediction Problem Dataset** with 614 loans, and I built it in Python using pandas and scikit-learn. I filled the missing values (the most common value for text-like columns and the median for the loan amount) and encoded the text columns as numbers.

> The data file is not included in this repository. It can be downloaded from Kaggle (search for "Loan Prediction Problem Dataset"). I used the training file, which contains the loan status.

What I did

1. **Explored the data:** 614 rows and 13 columns. About 69% of the loans were approved and 31% were not, so accuracy alone would be misleading (a model that always said "approved" would be about 69% accurate).
2. **Cleaned the data:** several columns had missing values (the most were in `Credit_History`, 50 rows). Dropping every row with a missing value would have removed 134 rows (about 22% of the data), so I filled the gaps instead. I also turned `Dependents` ("3+") into numbers and removed `Loan_ID`.
3. **Encoded the text columns** as numbers (Loan_Status as 1/0, the others with one-hot encoding).
4. **Split the data** into 80% for training and 20% for testing, keeping the same share of approved and rejected loans in both sets (`stratify`).
5. **Trained two models** (Logistic Regression and a balanced version) and compared them.
6. **Checked which clues matter most** and confirmed the main pattern in the data.

Key findings

Applicants whose credit history meets the guidelines are approved 79.0% of the time, against 7.9% for applicants whose credit history does not. The model relied most on credit history, and the other clues mattered much less.

Model results

I compared two models. The first Logistic Regression wrongly approved 16 of the 38 loans that should have been rejected. The balanced version wrongly approved 12, but it also wrongly rejected 10 good loans, against 2 for the first model.

Model (123 test loans)	Accuracy	Recall for loans to reject	Bad loans wrongly approved	Good loans wrongly rejected
Logistic Regression	0.85	0.58	16	2
Logistic Regression (balanced)	0.82	0.68	12	10

Which model is better depends on the lender: the balanced model avoids more bad loans, but it turns away more good customers.

Limitations

1. The dataset is small (614 loans, and only 123 in the test set), so the results could change with a different split.
2. I filled the missing values with guesses, for example the most common value for credit history, so the real approval rate for those applicants could be a bit different.
3. The model used details such as gender and marital status, and a real lender would have to think carefully about fairness before using them.